MIL-C-13573D
24 August 1983
SUPERSEDING
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8 March 1976

MILITARY SPECIFICATION

CALCIUM CHLORIDE, ANHYDROUS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of anhydrous calcium chloride.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	-	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter
PPP-F-320	-	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-TSC-S(A), Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-650	-	Explosive: Sampling, Inspection and Testing
MIL-STD-1168	-	Ammunition Lot Numbering

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM STANDARDS

D1193	-	Reagent Water
D3951	-	Commercial Packaging
E11	-	Wire-Cloth Sieves For Testing Purposes
E449	-	Analysis of Calcium Chloride

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Chemical characteristics. Calcium chloride shall conform to the chemical characteristics of table I when tested as specified therein.

TABLE I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Calcium chloride content	95	---	4.2.4.1
Volatile matter	---	4.0	4.2.4.2
Acidity (as HCl)	---	0.005	4.2.4.3
Alkalinity [as $Ca(OH)_2$]	---	0.3	4.2.4.3

3.2 Grit or insoluble particles. No more than 5 particles shall be retained on a 250-micrometer sieve and no particles shall be retained on a 425-micrometer sieve when tested as specified in 4.2.4.4.

3.3 Granulation. A minimum of 100 percent by weight shall pass through a 1.70-millimeter (mm) sieve when tested as specified in 4.2.4.5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the calcium chloride produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3). Each lot shall be identified and controlled in accordance with MIL-STD-1168.

4.2.2 Sampling. See 6.5 for sampling and testing precautions.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

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4.2.2.2 For calcium chloride test. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 250 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container form which it was taken.

TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
3 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure. See 6.5 for sampling and testing precautions.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample unit and shipping containers shall be examined for the following defects:

AQL 1.0 percent defective

(a) Container damaged or leaking

AQL 2.5 percent defective

(b) Contents per container not as specified
(c) Container not as specified
(d) Container closure not as specified
(e) Fiberboard pads, partitions, or liners missing or not as specified
(f) Unitization not as specified
(g) Marking incorrect, missing, or illegible

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. See 6.5 for sampling and testing precautions. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Calcium chloride content. Determine percent by weight calcium chloride in the specimen in accordance with the calcium chloride test procedure in ASTM E449.

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4.2.4.2 Volatile matter. Transfer approximately 5 g of the specimen into a tared glass-stoppered weighing bottle and weigh to the nearest 0.1 milligram (mg). Remove the stopper and place the bottle and contents in an oven maintained at 200° to 205°C for 2 hours. Replace the stopper, cool to room temperature in a desiccator, and weigh to the nearest 0.1 mg. Calculate the percent by weight volatile matter as follows:

$$\text{Percent volatile matter} = \frac{100\ (A - B)}{W}$$

where: A = Weight of bottle and contents before heating, in grams,
B = Weight of bottle and contents after heating, in grams, and
W = Weight of specimen in grams.

4.2.4.3 Acidity or alkalinity. Weigh to the nearest 0.01 g approximately 5 g of the specimen, which has been previously dried for 2 hours at 200° to 205°C, and dissolve in 50 milliliters (mL) of water. Add a few drops of phenolphthalein indicator solution. If the solution remains colorless, titrate it with 0.1N sodium hydroxide solution and calculate the percent by weight acidity as follows:

$$\text{Percent acidity} = \frac{3.647\ AB}{W}$$

where: A = Milliliters of sodium hydroxide solution used,
B = Normality of sodium hydroxide solution, and
W = Weight of dried specimen in grams.

If the solution turns red, titrate it with 0.1N hydrochloric acid and calculate the percent by weight alkalinity as follows:

$$\text{Percent alkalinity} = \frac{3.705\ AB}{W}$$

where: A = Milliliters of hydrochloric acid used,
B = Normality of hydrochloric acid, and
W = Weight of dried specimen in grams.

4.2.4.4 Grit or insoluble particles. Determine the grit or insoluble particles in the specimen in accordance with MIL-STD-650, method 106.1, using 200 mL of water plus 5 mL of 1 to 1 hydrochloric acid as the solvent.

4.2.4.5 Granulation. Weigh to the nearest 0.01 g approximately 25 g of the specimen, which has previously dried for 2 hours at 200° to 205°C, and place it on a 1.70-mm sieve conforming to ASTM E11. Tape a bottom pan under the sieve and cover with a tightly fitting cover. Shake for 10 minutes by hand or for 5 minutes in a mechanical shaker geared to produce 300 $\pm$ 15 gyrations and 150 $\pm$ 10 taps of the striker per minute. Weigh any material retained on the sieve as quickly as possible and calculate the percent by weight passing.

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5. PACKAGING

5.1 Unit packing. Calcium chloride shall be unit packed level A, B, or industrial as specified (see 6.2).

5.1.1 Level A. A quantity of 5 pounds (+1 or -0 ounce) of calcium chloride shall be unit packed in a metal can conforming to type V (round), class 2 of PPP-C-96 with exterior coating plan B with side seam striped.

5.1.2 Level B. A quantity of 5 pounds (+1 or -0 ounce) of calcium chloride shall be unit packed in a metal can conforming to type V (round), class 2 of PPP-C-96 with exterior coating plan A.

5.1.3 Industrial. A quantity of 5 pounds (+1 or -0 ounce) of calcium chloride shall be unit packed in a metal can in accordance with ASTM D3951.

5.2 Packing. Calcium chloride shall be packed level A, B, or industrial as specified (see 6.2).

5.2.1 Level A. Twelve cans of calcium chloride unit packed as specified in 5.1.1 shall be packed in an upright position in three rows of four cans per row in a close-fitting wood box conforming to overseas type, style optional, grade A of PPP-B-601; class 2, style 4, grade A of PPP-B-621; or class 3, style optional of PPP-B-585. The wood parts of PPP-B-585 shall be preserved as specified for the grade A box of PPP-B-621. Each box shall be designed for a type 2 load. All inside faces of the box shall be lined with fiberboard. Each can shall be placed in a full-can-height, close-fitting cell formed by interlocking sheets of half-slotted paritions. Fiberboard pads shall be inserted in the box wherever needed to prevent motion of contents. Liners, partitions and pads shall be formed from fiberboard conforming to grade V3c of PPP-F-320. The box shall be closed and reinforced in accordance with the requirements in the applicable box specification.

5.2.2 Level B. Calcium chloride shall be packed level B in the same manner as specified for level A except that the box shall be fiberboard conforming to grade V3c of PPP-B-636.

5.2.3 Industrial. Twelve cans of calcium chloride shall be packed industrially in accordance with ASTM D3951.

5.3 Unitization. Uniform quantities of level A or B packs of calcium chloride shall be palletized in accordance with the applicable requirements for load type 1 of MIL-STD-147, using the softwood pallet conforming to type IV of NN-P-71. Industrial packs shall be unitized in a manner to assure carrier acceptance, conformance with Uniform Freight Classification Rules, National Motor Freight Classification Rules, and the rules and regulations applicable to any other intended mode of transportation, and to assure stability of loads stacked in storage.

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5.4 Marking. Level A and B containers and unitized loads shall be marked in accordance with MIL-STD-129. Industrial containers shall be marked in accordance with ASTM D3951. Unitized loads of industrial containers shall be marked to identify the load and assure safe and expedient arrival at destination. Each container shall be marked to show lot number and date of manufacture of the calcium chloride. In addition, each container shall be durably and legibly marked with the following precautionary marking:

WARNING: CAUSES EYE AND SKIN IRRITATION

HARMFUL IF SWALLOWED

Will burn skin and eyes.
Avoid contact with eyes, skin and clothing.
Wear rubber overclothing, including gloves.
Wash thoroughly after handling.
If swallowed will cause nausea and vomiting.
Avoid breathing dust.
Keep container closed.
Use with adequate ventilation.

FIRST AID: In case of contact, flush affected areas with plenty of water. If in eyes, hold eyelids open and flush with plenty of water for at least 15 minutes. Remove contaminated clothing and shoes. If swallowed and victim is concious, have victim drink water or milk. If swallowed and victim is unconscious or having convulsions do nothing except keep victim warm. Call for immediate medical attention. If inhaled, remove to fresh air, if discomfort persists, get medical attention.

6. NOTES

6.1 Intended use. Calcium chloride covered by this specification is intended for use as a dehydrating agent for aluminized high explosives.

6.2 Ordering data. Acquisition documents should specify the title, number, and date of this specification, and the level of unit packing and packing required (see 5.1 and 5.2).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

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6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

6.5 Sampling and testing precautions. This specification covers inspection of chemical material (Calcium Chloride, Anhydrous) which is potentially hazardous to personnel. All applicable safety rules, regulations, and procedures must be followed in the handling and processing of this material.

Custodians:

Army - EA
Navy - OS

Review activities:

Army - MD, AR
DLA - GS

Preparing activity:

Army - EA

Project No. 6810-B394

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